

Seongbin Lim, PhD

AI Researcher and Engineer

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SUMMARY

AI researcher and engineer specializing in LLM applications, agent systems, computer vision, and production machine learning. Experienced in leading research teams, shaping product direction, and taking models from research and prototyping through deployment. PhD in Computer Science with a research background in self-supervised learning for biomedical imaging.

EXPERIENCE

AI Scientist

Oct 2024 - Jun 2026

Mind AI | Seoul, South Korea

- Researched and prototyped methods to extend LLM capabilities through logical validation, prompt optimization, context engineering, and agent harness design.
- Proposed and designed a domain-specific language for agents, built on top of agent SDKs.
- Established engineering patterns for agent development and contributed technical direction to product and architecture decisions.
- Built proof-of-concept agent products in a fast-moving startup environment.
- Fine-tuned BERT models for natural-language data imputation and classification.
- Trained and served graph neural networks for heterogeneous inputs, managing the model development and deployment lifecycle.

Research Team Lead

Mar 2023 - Oct 2024

NoriSpace | Seoul, South Korea

- Led a five-person machine learning research and engineering team developing AI solutions for business process automation.
- Collaborated with Kyobo Life Insurance and Hyundai Marine & Fire Insurance on document fraud detection using image forensics and anomaly detection.
- Fine-tuned a locally deployed LLM and developed a retrieval-augmented generation application with DB Insurance.
- Developed and packaged computer vision models for deployment on customer-managed, on-premises infrastructure.
- Conducted a four-month research visit at Mila in Montréal, Canada.
- Presented the company's technology to investors and international audiences, including CES 2024 and MWC 2024.

Doctoral Researcher

Oct 2019 - Oct 2023

Laboratoire d'Optique et Biosciences, École Polytechnique | Palaiseau, France

- Developed NU-Net, a self-supervised convolutional neural network that enhances cells and nuclei across diverse biomedical imaging modalities.
- Trained and evaluated models on 25 datasets and released lightweight pretrained models as open-source software and a Napari plugin.

- Created BioImageLoader, a Python library providing unified data loading, augmentation, concatenation, and batching for heterogeneous bioimage datasets.
- Published research at the ICCV 2023 BioImage Computing workshop and collaborated on interdisciplinary microscopy research.

SELECTED PROJECTS

Agent Slash Commands: A collection of reusable slash-style skills and workflows for coding agents.

NU-Net: Open-source self-supervised image filter for enhancing cells and nuclei in bioimages, distributed as a Python package and Napari plugin. [Source](#) / [PyPI](#) / [Publication](#)

BioImageLoader: Python library for loading heterogeneous biomedical image datasets in machine learning workflows. [Source](#) / [PyPI](#) / [Preprint](#)

TECHNICAL SKILLS

AI / ML	LLM applications, AI agents, prompt and context engineering, RAG, computer vision, self-supervised learning, BERT, graph neural networks, model training and serving
Programming	Python, Bash, JavaScript, Go, R, MATLAB
Systems	Linux, on-premises model deployment, open-source software development

EDUCATION

PhD in Computer Science Institut Polytechnique de Paris / École Polytechnique, France Dissertation: <i>Versatile Machine Learning for Neurodevelopmental Imaging</i>	2023
MScEng in Innovation Technologique : Ingénierie et Entrepreneuriat École Polytechnique and Kyung Hee University, France and South Korea	2019
BSc in Information Display Engineering Kyung Hee University, Seoul, South Korea	2017

SELECTED PUBLICATIONS

1. **S. Lim**, E. Beaufepaire, and A. Chessel. "NU-Net: A Self-Supervised Smart Filter for Enhancing Blobs in Bioimages." BioImage Computing Workshop, ICCV, 2023. [Paper](#)
2. **S. Lim**, X. Zhang, E. Beaufepaire, and A. Chessel. "BioImageLoader: Easy Handling of Bioimage Datasets for Machine Learning." arXiv:2303.02158, 2023. [Preprint](#)
3. T. P. L. Ung, **S. Lim**, et al. "Simultaneous NAD(P)H and FAD Fluorescence Lifetime Microscopy of Long UVA-Induced Metabolic Stress in Reconstructed Human Skin." *Scientific Reports* 11, 22171, 2021. [Paper](#)

LANGUAGES

Korean (native) | English (fluent) | French (intermediate) | Japanese (basic)